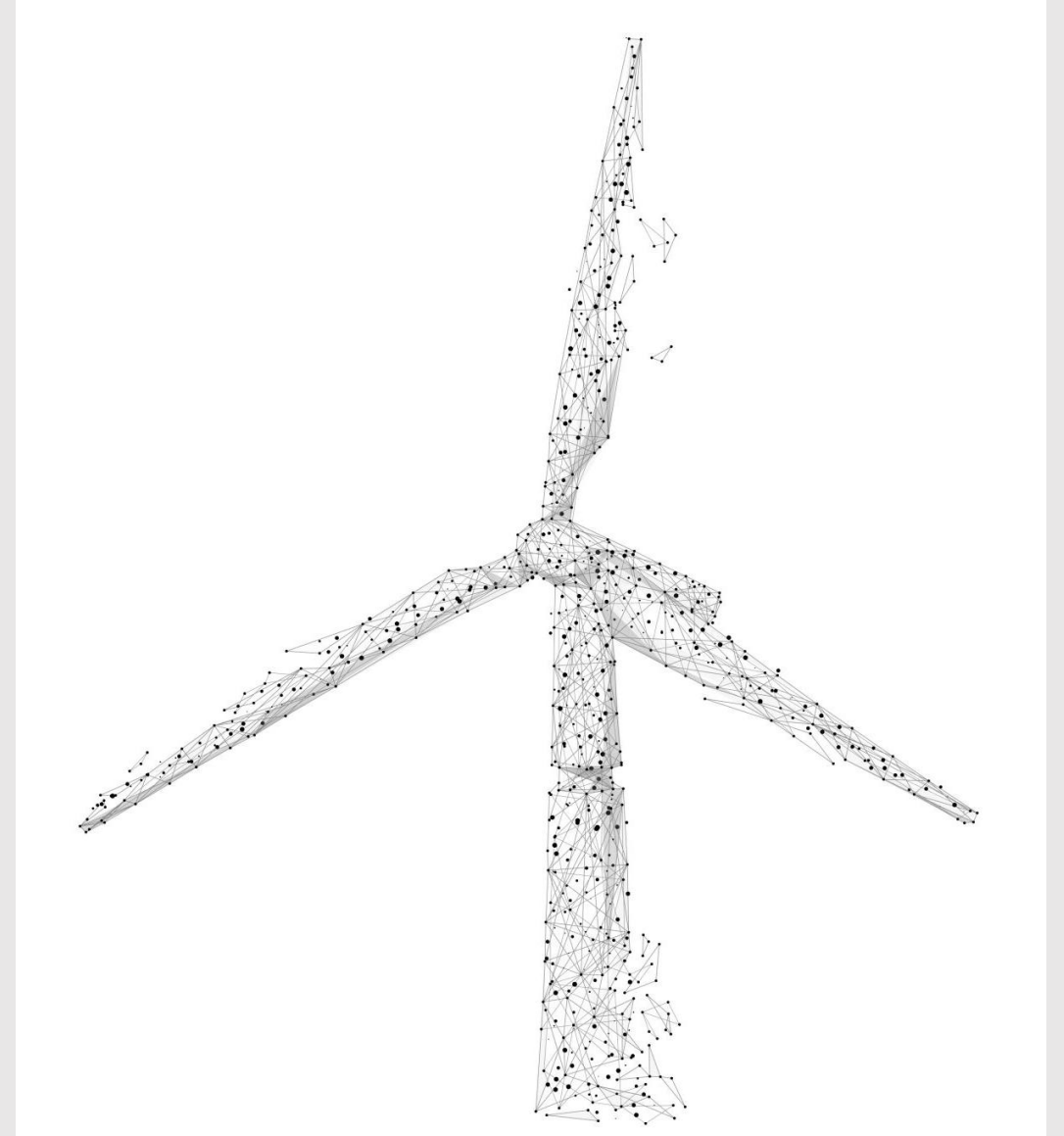

Wind Site: Phase 1

Name: Pingfan Teng

UUN: S2078106

Date: 05/03/2024



Wind Site: Phase 1

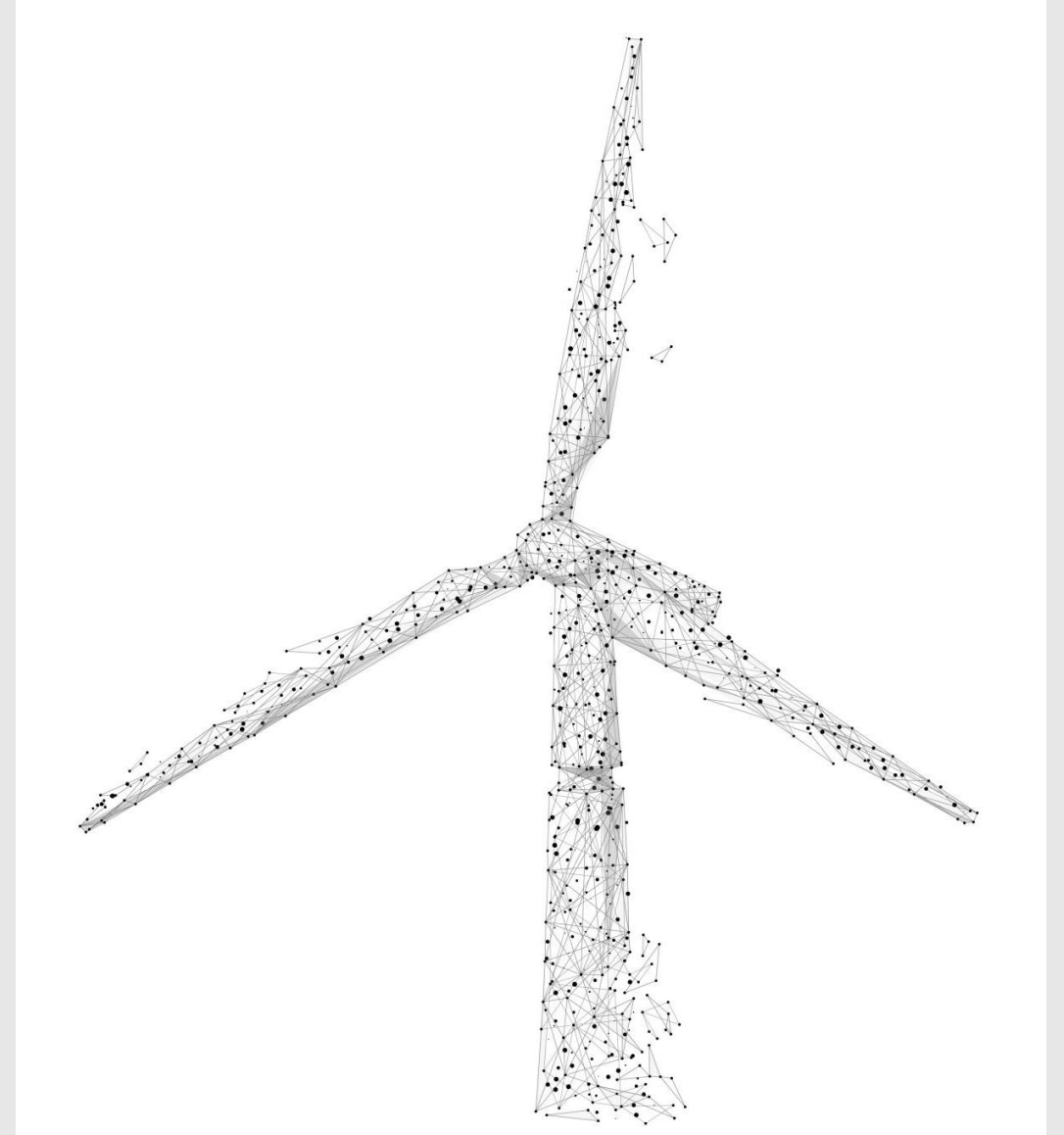
Completed Tasks:

Identified the IEC wind class on site

Selected the turbine

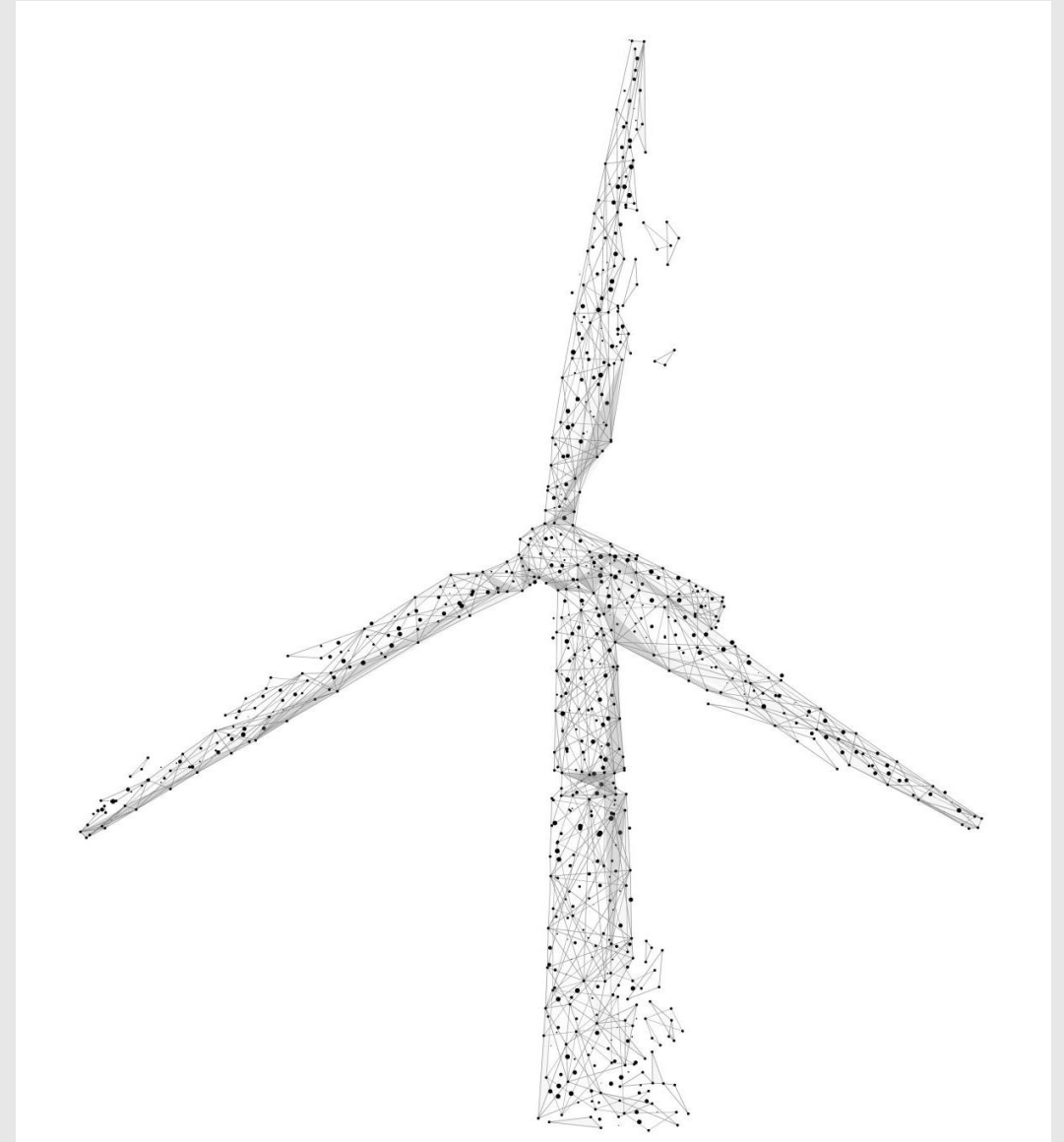
Estimated the energy yield with losses

Assessed the site (wake, shadow flicker, noise, social/environmental impacts)



Assumptions

- *Standard air density of 1.225 kg/m^3*
- *Only consider losses from noise/environmental curtailment, turbine performance and wake effect.*
- *Maximum tip height is 150 m*
- *Filter $\leq 3 \text{ m/s}$ from the nacelle mounted anemometer wind speed data*

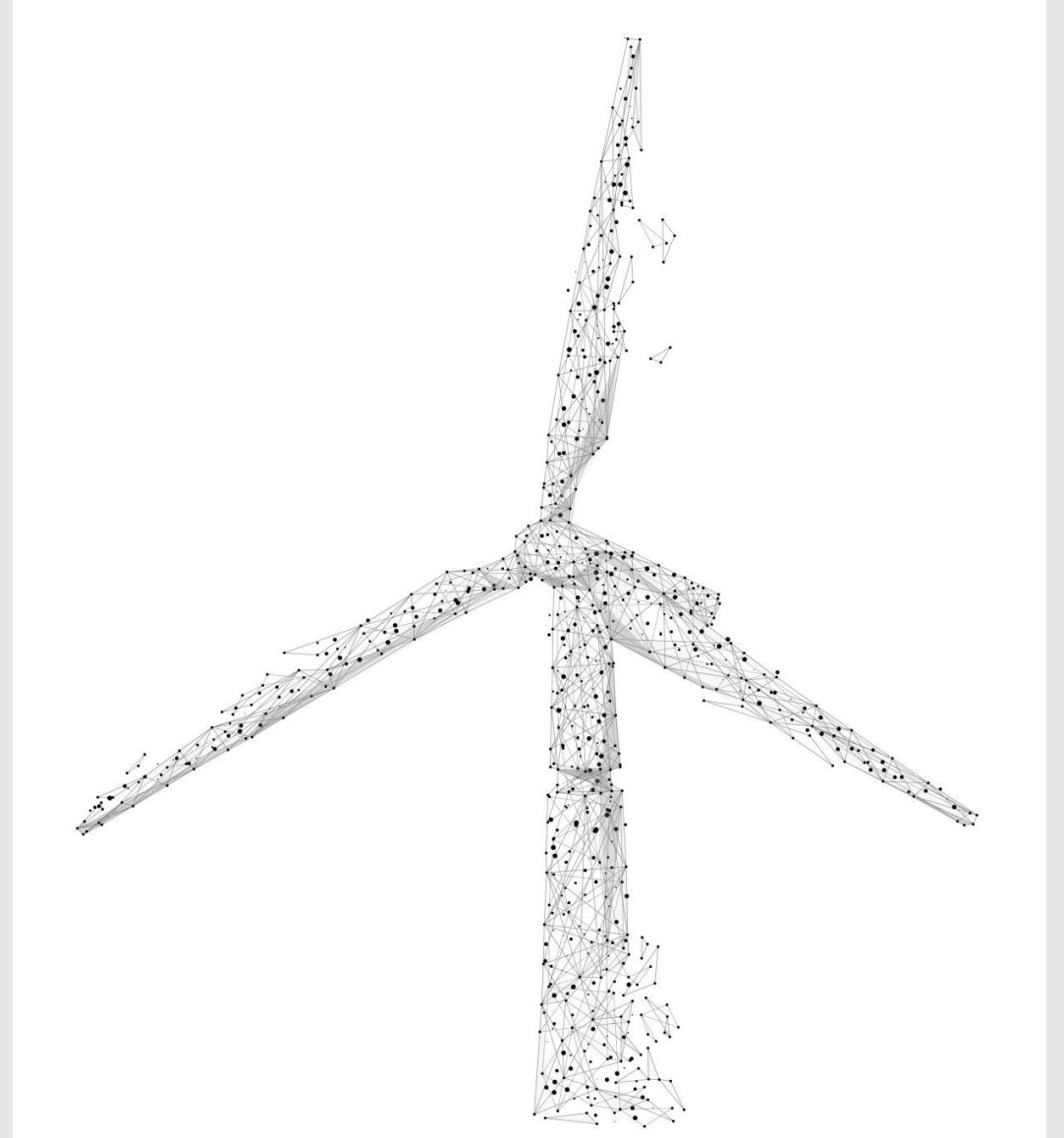


Uncertainties

There were uncertainties found in the calculations of wind speed, but those were insignificant based on the confidence level method.

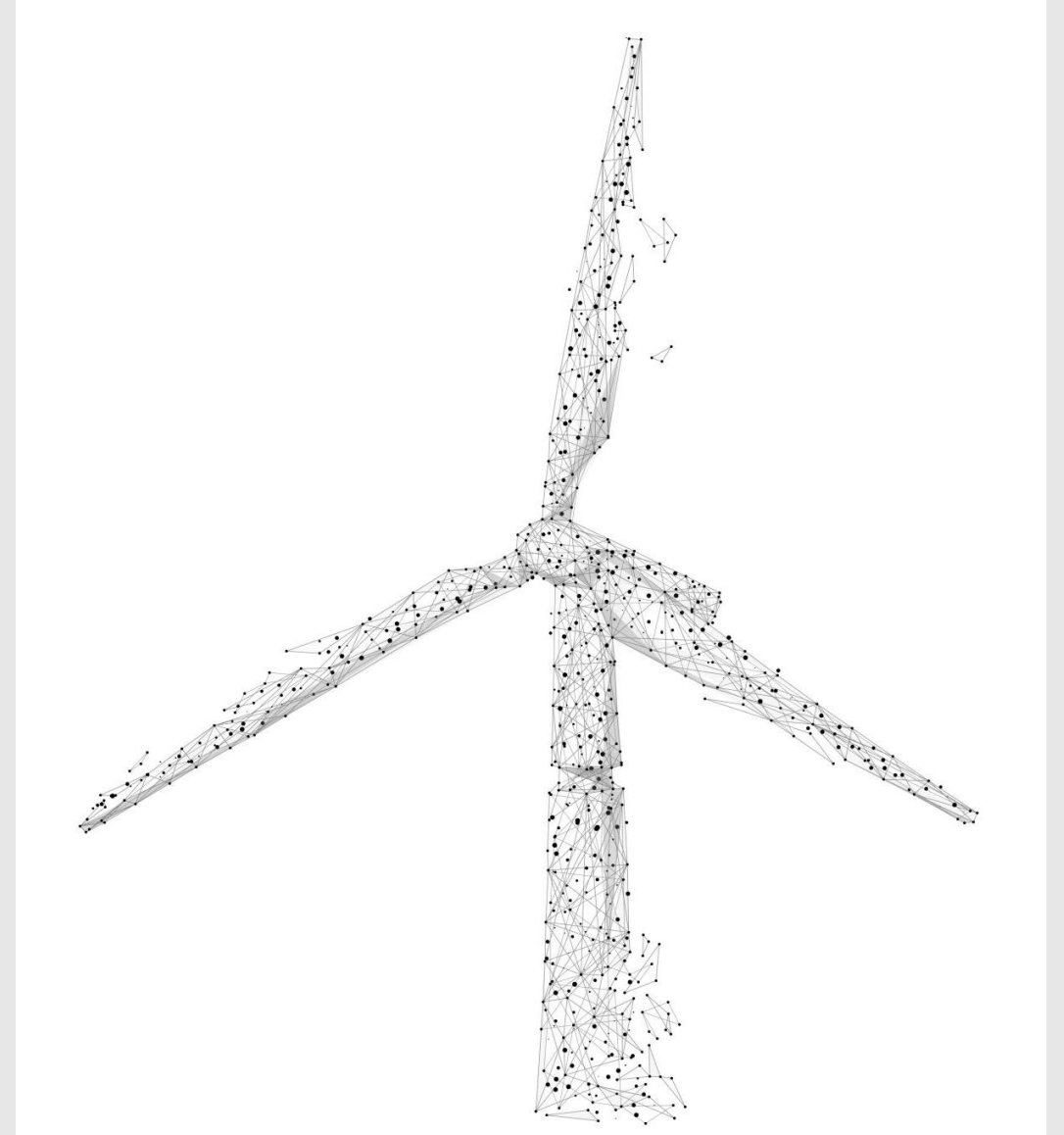
Furter uncertainties were found in the MCP, this is due to the fact that the met data sets are discrete values, but the ones at Penmanshiel are not.

Furtermore, uncertainties in energy loss due to wake loss, but were less than 1%.

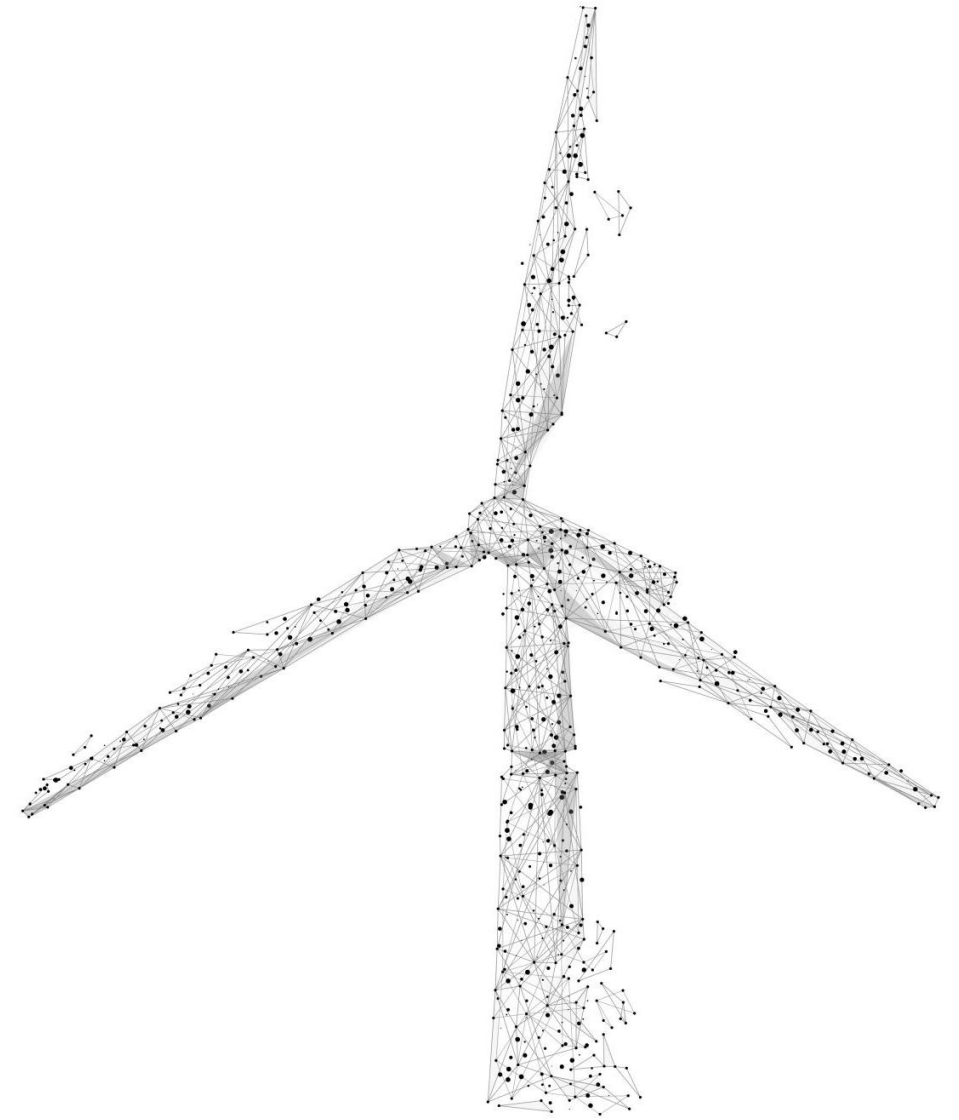
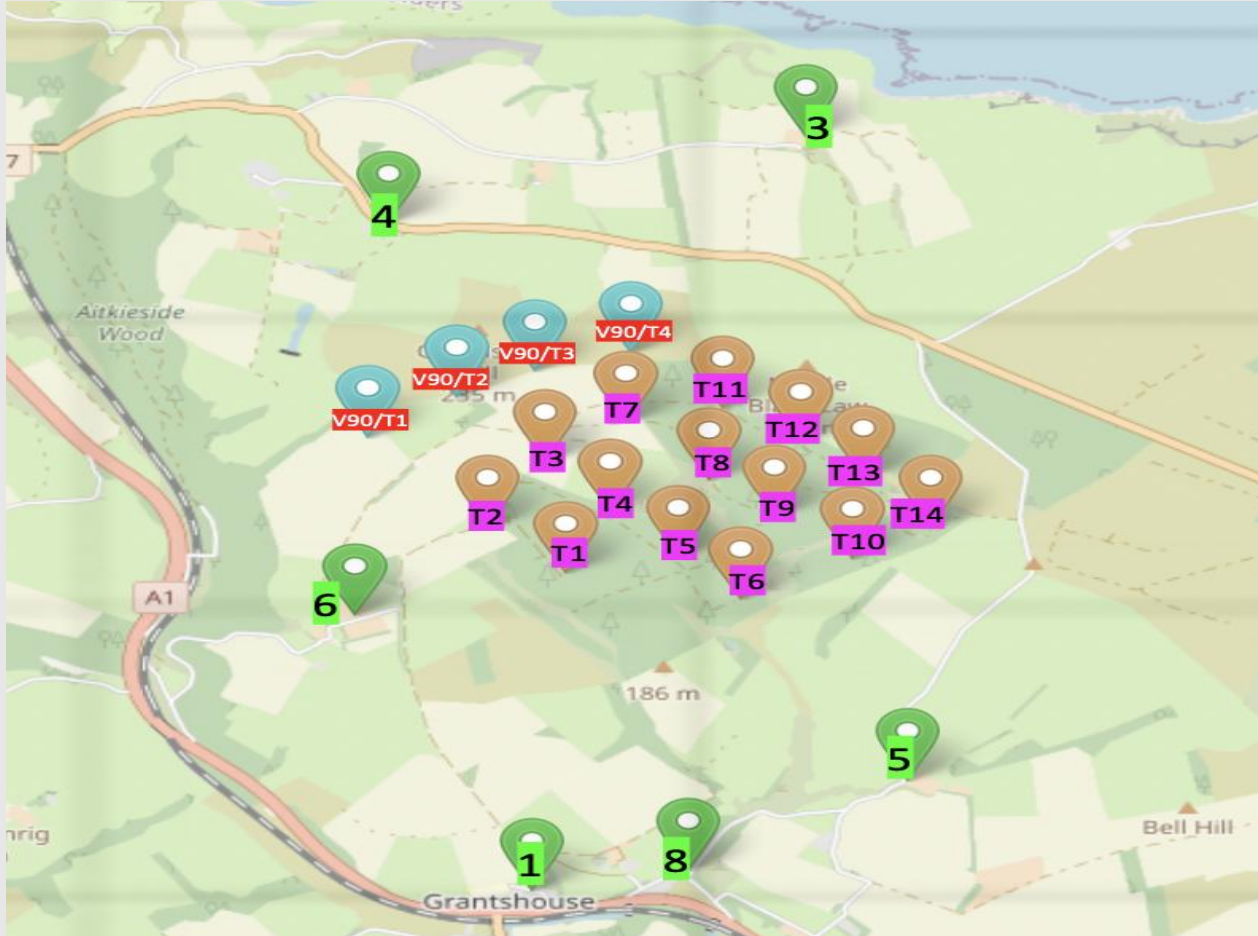


Turbine Selection

- *Selected turbine is V90/3.0 with 4 sound mode, manufactured by Vestas*
- *Adding up to 12 MW capacity*



Turbine Layout



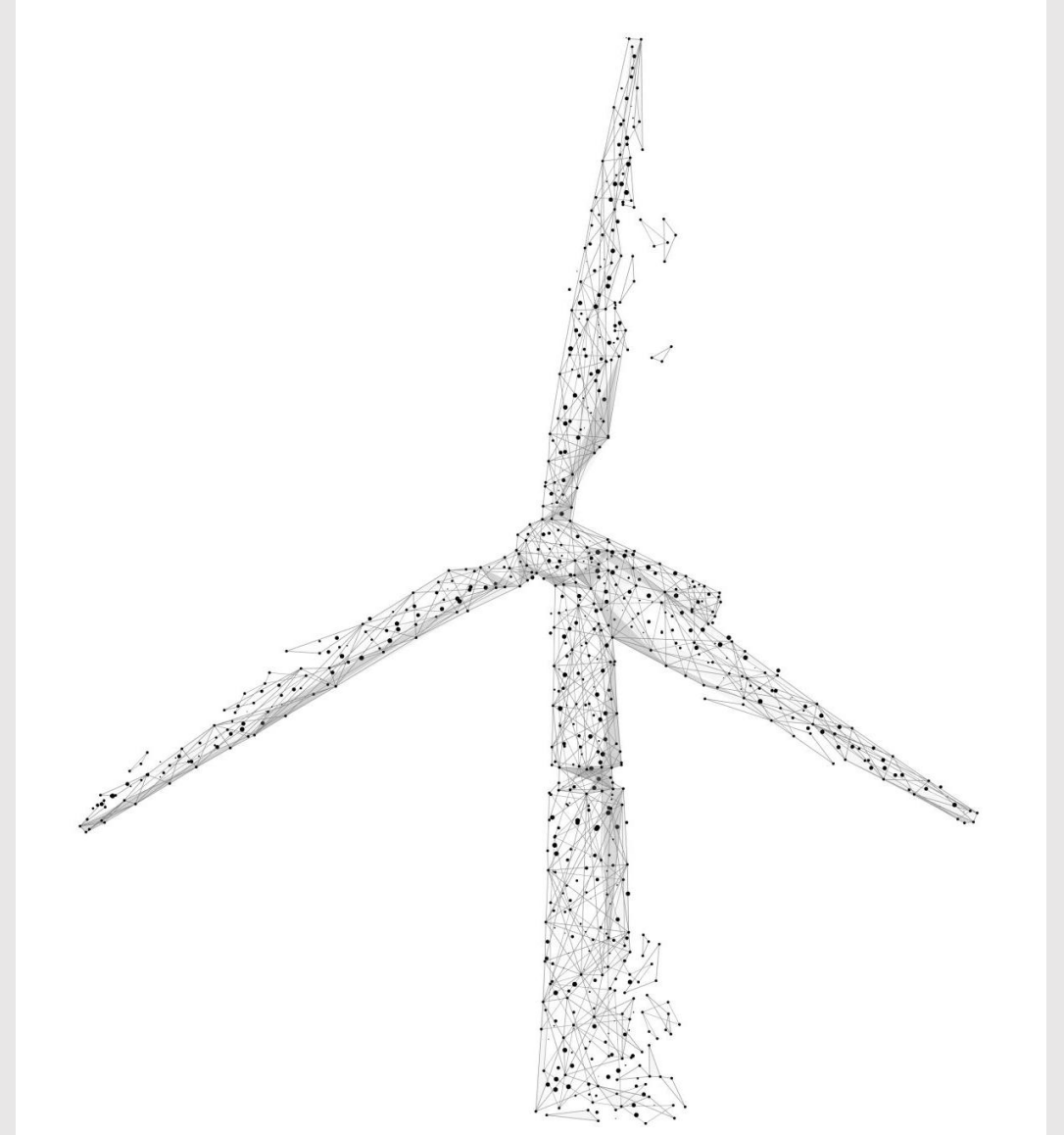
Produced energy

P50 energy yield of the 4 new installed turbines of P50 is 11183 MWh, P90 is about 9861 MWh.

UK average household annual usage = 2700kWh,

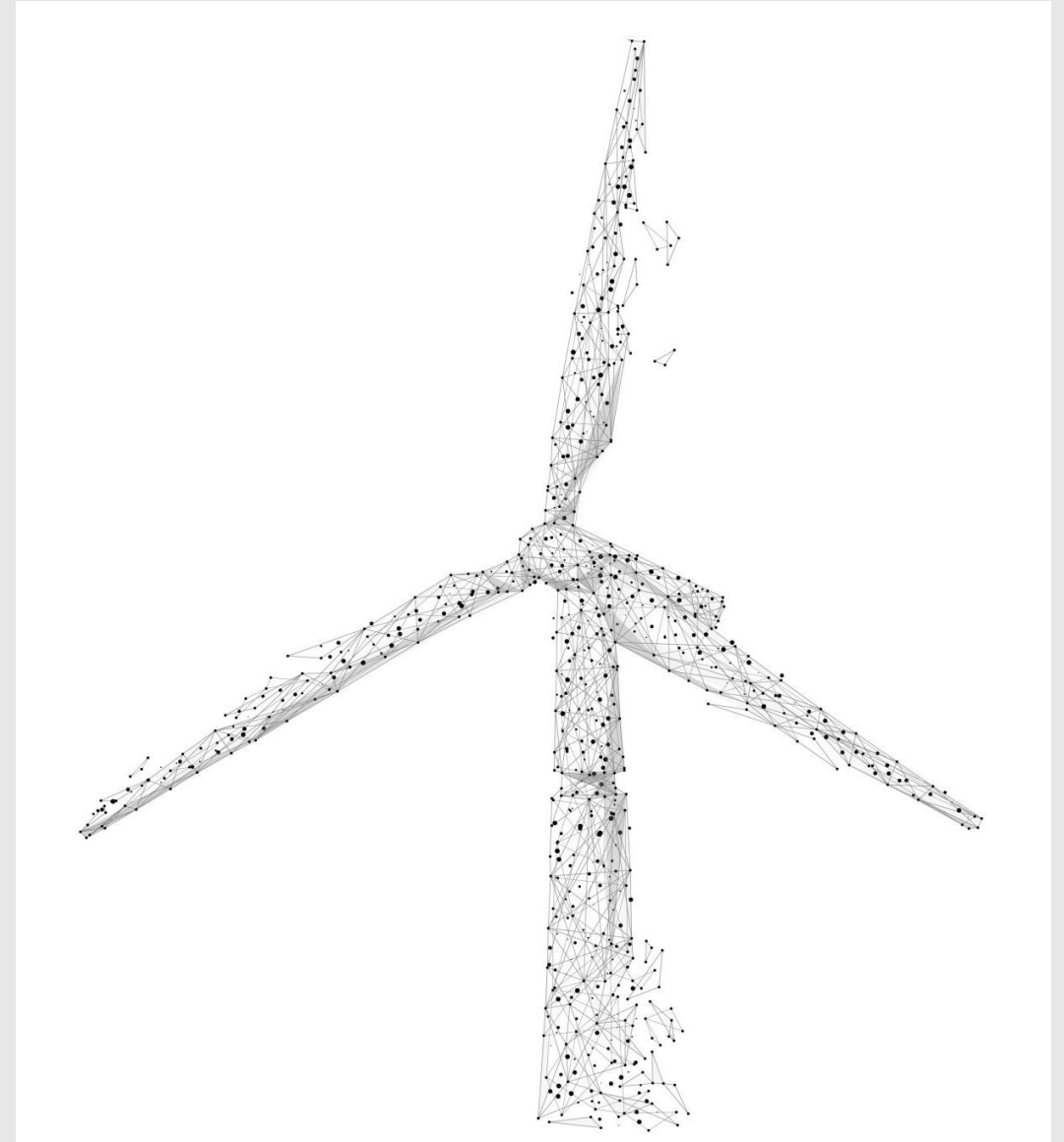
Hence, 90% of the year it can power up 3652 UK homes and 50% of the year it can power up 4141 homes.

(excluded old turbines at Penmanshiel Wind Farm)



Next steps?

- *Obtain more met station data from other met stations*
- *Turbine load analysis*
- *Analysis the wind class of the Drone hill wind farm*
- *Organise several meetings with the local communities and negotiate about the noise issues*
- *Conduct surveys and collect feedbacks*



Thank you for your patience

